

CRITICAL INFRASTRUCTURE MONITORING

{{device_name}} Monitoring Report

Device {{device_id}} — Site {{site_name}}

Project: {{project_name}}

Device ID: {{device_id}}

Site: {{site_name}}

Reporting period: {{reporting_period}}

Prepared by: {{author}}

Reviewed by: {{reviewer}}

Version: {{version}}

Status: {{overall_status}}

Agenda

1. Executive summary and overall status
2. Monitored asset description and operating context
3. Key performance indicators for the period
4. Alarm and incident analysis
5. Maintenance activities and interventions
6. Milestones and compliance tracking
7. Risk register for the monitored asset
8. Recommendations and actions for the next period
9. Appendices

This agenda is fixed by the monitoring service contract {{contract_ref}} and cannot be shortened without the asset owner's written approval.

Document Control

Field	Value
Project	{{project_name}}
Device	{{device_name}} {{device_id}}
Site	{{site_name}}
Reporting period	{{reporting_period}}
Prepared by	{{author}}
Reviewed by	{{reviewer}}
Contract reference	{{contract_ref}}
Version	{{version}}

Distribution List

Name	Role	Organisation
{{owner_rep}}	Asset owner representative	{{owner_org}}
{{ops_manager}}	Operations manager	{{operator_org}}
{{author}}	Monitoring engineer	{{company_name}}
{{reviewer}}	Quality reviewer	{{company_name}}

1. Executive Summary

During {{reporting_period}}, the device {{device_name}} ({{device_id}}) installed at {{site_name}} achieved an availability of {{availability_pct}} against a contractual target of {{availability_target}}. The overall status for the period is {{overall_status}}.

Summary item	Value
Overall status	{{overall_status}}
Availability	{{availability_pct}} (target {{availability_target}})
Critical alarms	{{critical_alarms}}
Major alarms	{{major_alarms}}
Minor alarms	{{minor_alarms}}
Corrective interventions	{{corrective_count}}
Preventive tasks completed	{{preventive_count}} / {{preventive_planned}}
Open risks at period end	{{open_risks}}

Table 1.1 — Period summary for {{device_id}}.

1.1 Headline Comment

"{{device_name}} operated within its design envelope for {{in_envelope_pct}} of the period. One critical alarm ({{critical_alarm_ref}}) required a site visit; root cause was identified and corrected within {{repair_duration}}." — {{author}}, monitoring engineer.

2. Monitored Asset Description

2.1 Identification

Attribute	Value
Device name	{{device_name}}
Device ID	{{{device_id}}}
Device class	{{device_class}}
Manufacturer	{{manufacturer}}
Serial number	{{serial_number}}
Firmware version	{{firmware_version}}
Installation date	{{install_date}}
Site	{{site_name}}
Site criticality tier	{{site_criticality}}
Network segment	{{network_segment}}

Table 2.1 — Device identification record.

2.2 Operating Context

{{device_name}} is part of the {{site_name}} critical infrastructure, classified as criticality tier {{site_criticality}} under the {{regulation_ref}} regulation. The device monitors {{monitored_parameter_list}} and reports telemetry every {{telemetry_interval}} to the central platform over the {{network_segment}} segment.

Loss of this device would degrade the site's protection function to {{degraded_mode}}. A backup channel is provided by {{backup_means}}.

2.3 Monitored Parameters

Parameter	Unit	Normal range	Alarm threshold	Critical threshold
{{param1_name}} }	{{param1_unit}} }	{{param1_normal}} }	{{param1_alarm}} }	{{param1_critical}} }
{{param2_name}} }	{{param2_unit}} }	{{param2_normal}} }	{{param2_alarm}} }	{{param2_critical}} }
{{param3_name}} }	{{param3_unit}} }	{{param3_normal}} }	{{param3_alarm}} }	{{param3_critical}} }
{{param4_name}} }	{{param4_unit}} }	{{param4_normal}} }	{{param4_alarm}} }	{{param4_critical}} }

Table 2.2 — Monitored parameters and thresholds configured for {{{device_id}}}.

3. Key Performance Indicators

The following KPIs are computed from raw telemetry for the reporting period ({{reporting_period}}). The measurement window is ({{measurement_window}}).

KPI	Unit	Target	Actual	Status
Availability	%	{{availability_target}}	{{availability_pct}}	{{kpi_avail_status}}
Telemetry completeness	%	{{kpi_tele_target}}	{{kpi_tele_actual}}	{{kpi_tele_status}}
Mean time between failures	days	{{kpi_mtbef_target}}	{{kpi_mtbef_actual}}	{{kpi_mtbef_status}}
Mean time to repair	h	{{kpi_mttr_target}}	{{kpi_mttr_actual}}	{{kpi_mttr_status}}
False alarm rate	%	{{kpi_false_target}}	{{kpi_false_actual}}	{{kpi_false_status}}
Data latency (p95)	s	{{kpi_lat_target}}	{{kpi_lat_actual}}	{{kpi_lat_status}}
Firmware currency	version	{{kpi_fw_target}}	{{kpi_fw_actual}}	{{kpi_fw_status}}

Table 3.1 — KPI values for ({{device_id}}) over ({{reporting_period}}).

3.1 Trend Commentary

Availability has remained above target for ({{consecutive_ok_periods}}) consecutive periods. Telemetry completeness dipped to ({{tele_dip_value}}) during the ({{tele_dip_cause}}) event; buffering on the device recovered all stored samples, so no data was permanently lost.

- The p95 data latency of ({{kpi_lat_actual}}) remains comfortably inside the ({{kpi_lat_target}}) budget.
- False alarm rate improved after the threshold recalibration performed in ({{recalibration_period}}).
- Firmware is ({{fw_gap}}) behind the latest qualified release; an update window is proposed in section 8.

4. Alarm and Incident Analysis

4.1 Alarm Register

Alarm ref	Date	Severity	Parameter	Duration	Acknowledged by
{{alarm1_ref}}	{{alarm1_date}}	Critical	{{alarm1_param}}	{{alarm1_duration}}	{{alarm1_ack}}
{{alarm2_ref}}	{{alarm2_date}}	Major	{{alarm2_param}}	{{alarm2_duration}}	{{alarm2_ack}}
{{alarm3_ref}}	{{alarm3_date}}	Major	{{alarm3_param}}	{{alarm3_duration}}	{{alarm3_ack}}
{{alarm4_ref}}	{{alarm4_date}}	Minor	{{alarm4_param}}	{{alarm4_duration}}	{{alarm4_ack}}
{{alarm5_ref}}	{{alarm5_date}}	Minor	{{alarm5_param}}	{{alarm5_duration}}	{{alarm5_ack}}

Table 4.1 — Alarms raised by {{device_id}} during {{reporting_period}}.

4.2 Critical Incident Report — {{critical_alarm_ref}}

On {{incident_date}}, the device raised critical alarm {{critical_alarm_ref}} on parameter {{incident_param}}. The value reached {{incident_peak}} against a critical threshold of {{incident_threshold}}. The duty engineer was paged at {{page_time}} and acknowledged within {{ack_delay}}.

4.2.1 Timeline

Time	Event
{{tl1_time}}	Alarm {{critical_alarm_ref}} raised by {{device_id}}
{{tl2_time}}	Duty engineer paged and acknowledged
{{tl3_time}}	Remote diagnostic session opened
{{tl4_time}}	Site technician dispatched
{{tl5_time}}	Root cause identified: {{root_cause}}
{{tl6_time}}	Corrective action completed, parameter back in normal range
{{tl7_time}}	Alarm cleared, incident closed

Table 4.2 — Incident timeline for {{critical_alarm_ref}}.

4.2.2 Root Cause and Corrective Action

Root cause: {{root_cause}}. The corrective action {{corrective_action}} restored normal operation within {{repair_duration}}. A preventive measure {{preventive_action}} was added to the maintenance plan to avoid recurrence.

4.3 Alarm Statistics

Severity	Count	Share of total	Avg. acknowledgment time	Avg. resolution time
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Critical	{{critical_alarms}}	{{critical_share}}	{{crit_ack_avg}}	{{crit_res_avg}}
Major	{{major_alarms}}	{{major_share}}	{{maj_ack_avg}}	{{maj_res_avg}}
Minor	{{minor_alarms}}	{{minor_share}}	{{min_ack_avg}}	{{min_res_avg}}

Table 4.3 — Alarm statistics for the period.

5. Maintenance Activities

5.1 Preventive Maintenance

Task	Planned date	Executed date	Result	Technician
Sensor calibration check	{{pm1_plan}}	{{pm1_exec}}	Pass	{{pm1_tech}}
Backup battery load test	{{pm2_plan}}	{{pm2_exec}}	Pass	{{pm2_tech}}
Communication link failover test	{{pm3_plan}}	{{pm3_exec}}	Pass	{{pm3_tech}}
Firmware health audit	{{pm4_plan}}	{{pm4_exec}}	Pass with remark	{{pm4_tech}}
Cabinet thermal inspection	{{pm5_plan}}	{{pm5_exec}}	Pass	{{pm5_tech}}

Table 5.1 — Preventive maintenance tasks completed during ({{reporting_period}}) ({{preventive_count}} of ({{preventive_planned}}) planned tasks).

5.2 Corrective Interventions

Ref	Date	Linked alarm	Action	Downtime
{{ci1_ref}}	{{ci1_date}}	{{critical_alarm_ref}}	{{ci1_action}}	{{ci1_downtime}}
{{ci2_ref}}	{{ci2_date}}	{{alarm3_ref}}	{{ci2_action}}	{{ci2_downtime}}

Table 5.2 — Corrective interventions (total ({{corrective_count}}) for the period).

5.3 Spare Parts Consumed

Part reference	Description	Quantity	Stock after use
{{part1_ref}}	{{part1_desc}}	{{part1_qty}}	{{part1_stock}}
{{part2_ref}}	{{part2_desc}}	{{part2_qty}}	{{part2_stock}}

Table 5.3 — Spare parts consumed on ({{device_id}}) during the period.

6. Milestones and Compliance Tracking

6.1 Contractual Milestones

Milestone	Due date	Status	Evidence
Quarterly monitoring report delivered	{{ms1_due}}	Done	{{ms1_evidence}}
Annual sensor recalibration	{{ms2_due}}	Scheduled	—
Disaster-recovery drill participation	{{ms3_due}}	Done	{{ms3_evidence}}
Firmware qualification review	{{ms4_due}}	In progress	—

Table 6.1 — Milestones under contract {{contract_ref}}.

6.2 Regulatory Compliance Checklist

Requirement	Reference	Status	Comment
Continuous monitoring of critical parameters	{{regulation_ref}} art. {{reg_art1}}	Compliant	Coverage {{kpi_tele_actual}}
Alarm acknowledgment within {{sla_ack}}	Contract {{contract_ref}} annex B	Compliant	Avg {{crit_ack_avg}}
Data retention of {{retention_period}}	{{regulation_ref}} art. {{reg_art2}}	Compliant	Archive verified {{archive_check_date}}
Annual penetration test of telemetry path	Security policy {{sec_policy_ref}}	Compliant	Last test {{pentest_date}}
Backup communication channel	Contract {{contract_ref}} annex C	Compliant	Failover tested {{pm3_exec}}

Table 6.2 — Compliance checklist for {{device_id}}.

7. Risk Register

Risks below are specific to {{device_name}} ({{device_id}}) and its operating environment at {{site_name}}. The register is reviewed each reporting period by {{reviewer}}.

ID	Risk	Likelihood	Impact	Rating	Mitigation	Owner	Status
R-01	Single communication path during {{backup_means}} maintenance windows	Medium	High	{{r1_rating}}	{{r1_mitigation}}	{{r1_owner}}	Open
R-02	Firmware {{firmware_version}} approaching end of support ({{fw_eos_date}})	High	Medium	{{r2_rating}}	{{r2_mitigation}}	{{r2_owner}}	Open
R-03	Sensor drift beyond calibration interval	Low	Medium	{{r3_rating}}	{{r3_mitigation}}	{{r3_owner}}	Monitored
R-04	Unauthorised physical access to outdoor cabinet	Low	High	{{r4_rating}}	{{r4_mitigation}}	{{r4_owner}}	Monitored
R-05	Spare part {{part1_ref}} long lead time ({{part1_leadtime}})	Medium	Medium	{{r5_rating}}	{{r5_mitigation}}	{{r5_owner}}	Open

Table 7.1 — Risk register for {{device_id}} at end of {{reporting_period}} ({{open_risks}} open risks).

7.1 Risk Evolution

Compared with the previous period, risk R-02 increased from Medium to High likelihood because the vendor published the end-of-support notice for firmware {{firmware_version}}. All other ratings are stable. No risk was closed during the period.

8. Recommendations and Next Actions

#	Action	Priority	Owner	Due date
1	Schedule firmware update to {{kpi_fw_target}} during the {{fw_window}} maintenance window	High	{{r2_owner}}	{{action1_due}}
2	Order one additional {{part1_ref}} spare to cover the {{part1_leadtime}} lead time	High	{{r5_owner}}	{{action2_due}}
3	Rehearse degraded-mode procedure with the {{operator_org}} control room	Medium	{{ops_manager}}	{{action3_due}}
4	Extend threshold recalibration to the redundant sensor channel	Medium	{{author}}	{{action4_due}}
5	Update the site access register for the outdoor cabinet	Low	{{r4_owner}}	{{action5_due}}

Table 8.1 — Recommended actions for the next reporting period.

Completion of these actions will be verified in the next monitoring report covering {{next_period}}.

Appendix A — Raw KPI Computation Notes

Availability is computed as (period duration – unplanned downtime) / period duration, excluding approved maintenance windows. Telemetry completeness is the ratio of received samples to expected samples at the configured interval of {{telemetry_interval}}. Latency is measured platform-side as ingest timestamp minus device timestamp.

- Measurement window: {{measurement_window}}.
- Excluded maintenance windows: {{excluded_windows}}.
- Telemetry platform version: {{platform_version}}.
- Computation job reference: {{computation_job}}.

Appendix B — Variable Reference

Placeholder	Description	Example
{{project_name}}	Monitoring project or contract name	MERIDIAN Watch
{{device_name}}	Human name of the monitored device	Feeder Protection Relay FPR-7
{{device_id}}	Unique device identifier	CI-FPR-0007
{{site_name}}	Site where the device is installed	Vauchuse 225 kV Substation
{{reporting_period}}	Period covered by this report	Q2 2026
{{author}}	Monitoring engineer who wrote the report	L. Fabre
{{reviewer}}	Quality reviewer	M. Okonkwo
{{overall_status}}	RAG status for the period	GREEN

Table B.1 — Core variable reference (non-exhaustive).

Appendix C — Consolidated Event Log

The event log below lists every operational event recorded for {{device_id}} during {{reporting_period}}, as exported from the telemetry platform (job {{computation_job}}). Events are ordered chronologically; durations are expressed in hours:minutes.

Date/time	Event type	Severity	Description	Duration	Linked ref
{{ev1_time}}	Alarm raised	Critical	{{ev1_desc}}	{{ev1_dur}}	{{critical_alarm_ref}}
{{ev2_time}}	Alarm acknowledged	Critical	Acknowledged by {{alarm1_ack}}	—	{{critical_alarm_ref}}
{{ev3_time}}	Intervention started	Info	Technician {{pm1_tech}} on site	—	{{ci1_ref}}
{{ev4_time}}	Alarm cleared	Critical	Parameter back in normal range	{{ev4_dur}}	{{critical_alarm_ref}}
{{ev5_time}}	Alarm raised	Major	{{ev5_desc}}	{{ev5_dur}}	{{alarm2_ref}}

{{ev6_time}}	Communication failover	Warning	Primary link lost, backup {{backup_means}} active	{{ev6_dur}}	—
{{ev7_time}}	Maintenance task	Info	Backup battery load test — pass	—	{{pm2_exec}}
{{ev8_time}}	Alarm raised	Minor	{{ev8_desc}}	{{ev8_dur}}	{{alarm4_ref}}
{{ev9_time}}	Configuration change	Info	Threshold recalibration on {{param2_name}}	—	{{recalibration_period}}
{{ev10_time}}	Report generated	Info	Period export for {{reporting_period}}	—	{{computation_job}}

Table D.1 — Consolidated event log (fictional, template demonstration).

Appendix D — Approval

This report was prepared by {{author}} and reviewed by {{reviewer}}. Approval by the asset owner representative {{owner_rep}} is recorded in the contract management system under reference {{approval_ref}}.

Role	Name	Signature	Date
Prepared by	{{author}}		{{approval_date}}
Reviewed by	{{reviewer}}		{{approval_date}}
Approved by	{{owner_rep}}		{{approval_date}}

Table C.1 — Signature block.